Robot Introduction Lab

Prof. Oliver Kroemer

Introduction

In this lab, you will learn how to startup and shutdown the Franka Panda Robot, how to work safely with the Franka robot, and how to control the robot with frankapy. An overview of the robot components is shown if Fig. 1.

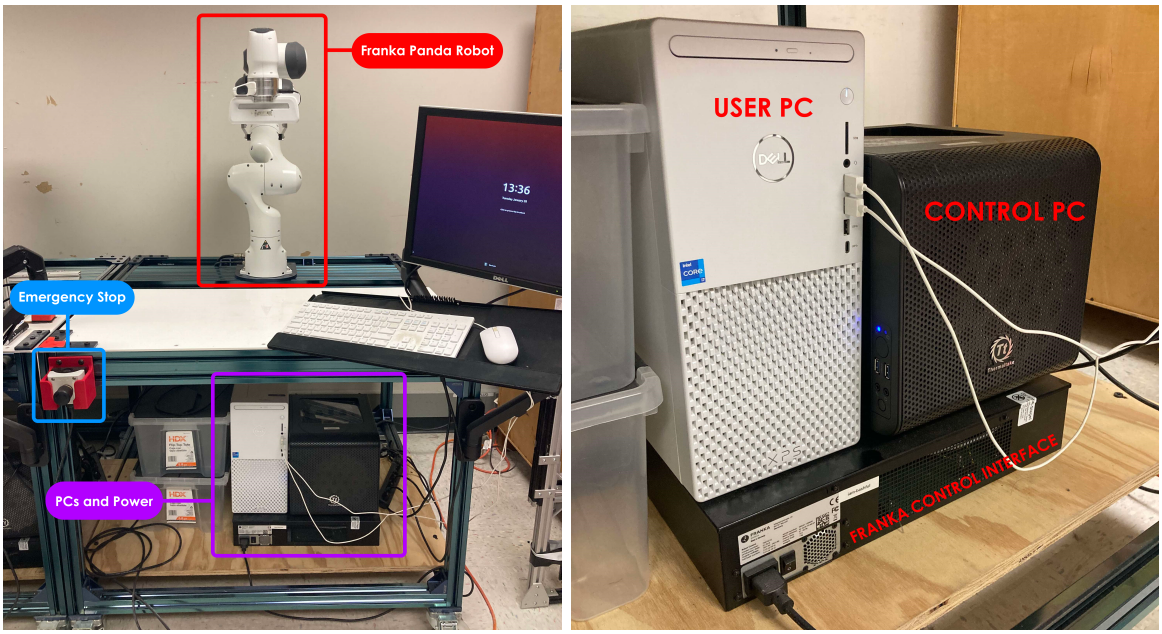

Figure 1: The different components of the Franka Setup in the lab.

Note the PC group (the right image). You have the User PC on the left, the Control PC on the right, and the Franka Control Interface on the bottom.

Note on Safety:

Safety is always of upmost importance when working with the robots. When running a program that commands the robot, please:

1. Stay outside of the robot's workspace (defined by the edges of the table).
2. Always keep a hand on the e-stop. Under no circumstances should the e-stop be out of reach.
3. The e-stop can be easily released by twisting it gently. Give it a try!

Important: You should never have to apply large forces to move the robot by hand. To avoid damage, forces should be similar to those of holding a cup of water.

Turning On and Unlocking the Robot

The following steps will allow you to turn on the robot and unlock its joints such that it can be used. To unlock the joints, you will need to SSH onto the control computer and access the robot's web interface.

1. Flip the `ON` switch on the Franka Control Interface (Flip the switch UP, so that the "line" is pressed in). Turn on both the control computer and the user computer. The light on the side panel of the Control PC should be lit blue when on. If it is not, then press the large circle button. Refer to Fig. 2.

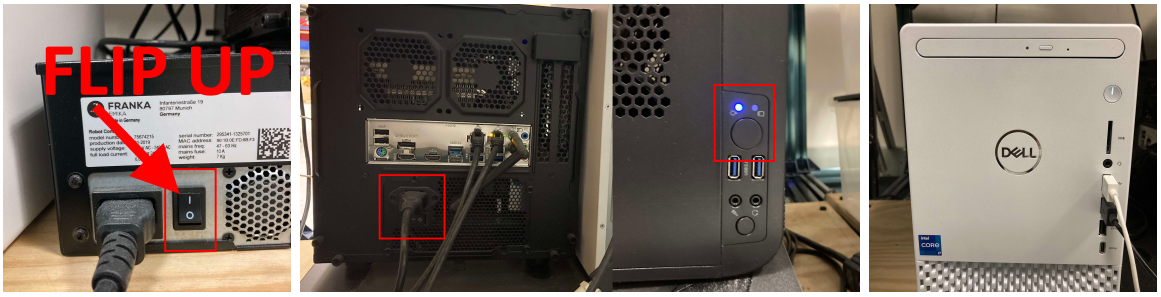

Figure 2: Left: Franka Control Interface; Middle: Control PC Power Buttons; Right: User PC

2. Log into the user computer **"student"** account with the password: **16-662**

3. SSH onto the control computer with using the following terminal command :
   `ssh -X student@iam-<name>`
   You will find the name on the top of the control computer's casing.

4. Launch Google Chrome by running `google-chrome` in the terminal.
   Troubleshooting if running google-chrome or -X is giving errors:
   (a) Try restarting the terminal
   (b) If google-chrome has an "instance is already running" error or similar, run this:
   `kill all google-chrome`
   (c) If connecting to the PC is not working, then safely turn off the PCs and try again.
   (d) If these don't help, then ask your TA for assistance.

5. Open the webpage **172.16.0.2** in the browser to access the web interface.
   (If it prompts for a login, the username is admin, and the password is shavethepandas)

6. Press "Click to unlock" as shown in Fig 3 to unlock the joints. Some PCs may have a newer Franka UI on Google Chrome. To unlock, search for the icons that look like a locked lock and unlocked lock. Click on the unlocked lock.
   Wait for the robot lights to turn blue. It is the state in which robot expects commands from the computer.

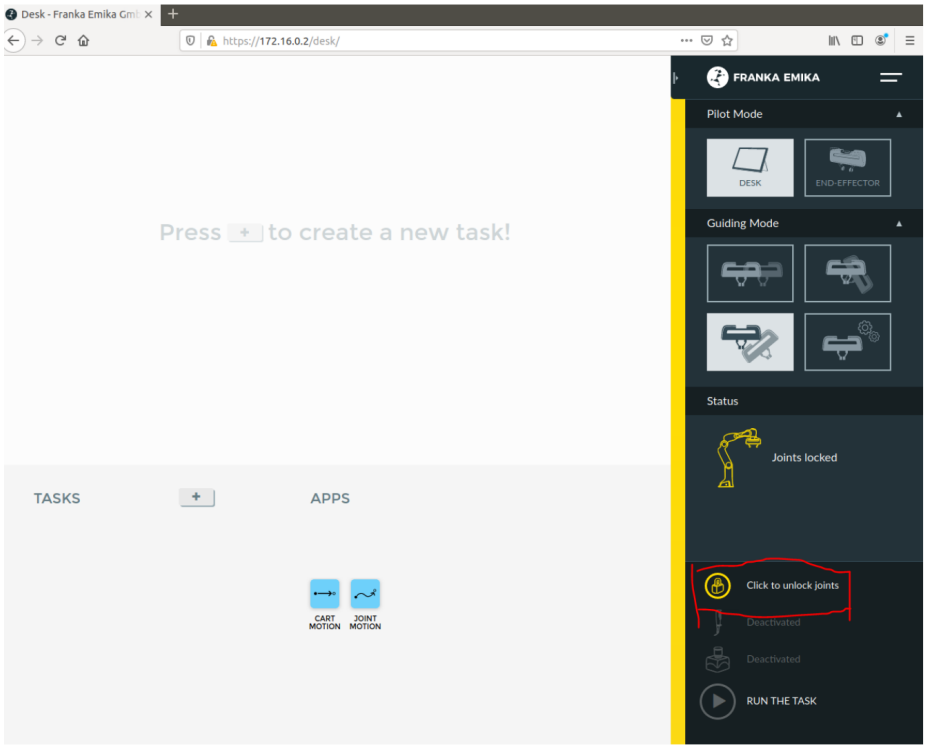

Figure 3: The location of the Unlock Joints button

Robot Indicator Lights (This is Information, not Instruction)

The robot has the following modes denoted by the indicator light on the side of the robot base:

- Yellow - Lock Mode. The robot's joints are locked, and it cannot be moved. Locking and unlocking the joints are done in a webpage on the Control PC (you will do this in a later section). Once the joints are unlocked, the robot can be either in White or Blue.
- White - Manual Mode. The robot may be moved by a human pressing down on the gray buttons near the robot hand. However, when the robot is in manual mode, it cannot be commanded via a program. White is achieved when the e-stop is pressed down.
- Blue - Program Mode. The robot can be commanded with a program, but it cannot be moved by a human pressing down on the gray buttons. To achieve Blue, twist and release the e-stop.
- Pink - Small error. Someone attempted to move the robot manually while it was in Blue. When in Pink, press down on the e-stop to return the robot to White.
- Red - Serious error. This happens when the robot incurs a significant collision. A full restart is required to move the robot again.

Moving the Robot by Hand

You will sometimes need to manually move the arm into a safe initial configuration, e.g., if you hit the emergency button while the arm was close to an obstacle or if the arm was in an awkward initial position. The following steps will allow you to enter gravity compensation mode to move the arm by hand.

1. Hit the emergency stop button. The robot lights should now turn white.
2. *Gently* squeeze the opposing grey buttons on the wrist flange. Note that these are deadman switches - if you press too firmly, it will think you are being electrocuted and stop. The arm should move with very little effort.
3. Bring the arm into a suitable starting configuration and release the wrist buttons.
4. Release the e-stop by twisting the e-stop button. The light should now be blue and ready for computer commands.

Starting the Control PC Server

We will be using the frankapy interface to control the robot arm. The interface uses a server on the Control PC to run skills at 1kHz. The server is started from a script on the user PC.

1. Open a new terminal, i.e., *not* the one you used to SSH onto the Control PC.
2. Make sure the docker repository 16662_RobotAutonomy is cloned at `/home/student`.
3. Make a new folder with your team name or number in `16662_RobotAutonomy/src/devel_packages`. Use this folder to store all the code for your team.
4. In a new terminal, run `roscore` to start the roscore
5. Run the start_control_pc script from the frankapy package
   `cd /home/student/Documents/frankapy`
   `bash ./bash_scripts/start_control_pc.sh -u student -i [control-pc-name]`
   This should launch 3 terminals which sets up frankapy to communicate with the robot. To reset frankapy communication, just kill the 3 terminals and rerun the script.
6. **Running the Docker Container:**
   - To run the Docker Container in a new terminal:
     `cd 16662_RobotAutonomy`
     `bash run_docker.sh`
   - Attach a terminal connected to the Docker Container:
     `bash terminal_docker.sh`

   You will use the docker docker to run your code for this lab.

NOTE: Whenever you hit the e-stop, you will need to restart the Control PC. Close the terminals and rerun the bash script.

Running a Simple (Reset Arm) Script

We will now run a simple script for resetting the arm. We assume that the control PC server setup is complete.

1. In a terminal attached to the docker container, run `python3 guide_mode.py`. The script puts the robot in guide mode, and you can now move the robot in any pose manually. **You *DO NOT* need to press the wrist buttons to move the robot in this state.**

2. After you have moved the robot to a random position, hit 4 and press enter. After a short lag, the robot should reset to the home position with its hand hovering above the table near the middle of its workspace.

Creating a Script to Move the Robot Arm

The next step will be to create a script for getting readings and sending commands to the robot arm. You should be able to run the script at each step to test the outcome.

1. Create your own project folder within the `~/16662_RobotAutonomy/src/devel_packages` folder. Create a Python script in the folder and open it for editing.

2. Setup a basic script as follows:
```
import numpy as np
from autolab_core import RigidTransform
from frankapy import FrankaArm
if __name__ == "__main__":
       fa = FrankaArm()
       fa.reset_joints()
```
This script will create a FrankaArm instance and use it to reset the joints. It is good practice to start with a reset command so that the arm always starts in the same configuration.

3. Add some functions to read out the robot's state:

   - Endeffector pose:
   ```
   T_ee_world = fa.get_pose()
   print('Translation:  {}'.format(T_ee_world.translation))
   print('Rotationn:  {}'.format(T_ee_world.quaternion))
   ```
   - Joint angles:
   ```
   joints = fa.get_joints()
   print('Joints:  {}'.format(joints))
   ```
   - Gripper width:
   ```
   gripper_width = fa.get_gripper_width()
   print('Gripper width:  {}'.format(gripper_width))
   ```
   - Endeffector forces:
   ```
   force_torque = fa.get_ee_force_torque()
   print('Forces and Torques:  {}'.format(force_torque))
   ```
   How much force is it sensing in the vertical direction?

4. Add a function call to move the robot to a desired joint configuration

- Arm:
  `fa.goto_joints([0.0, -0.7, 0.0, -2.15, 0.0, 1.57, 0.7])`
- Gripper:
  `fa.close_gripper()`
  `fa.open_gripper()`
  `fa.goto_gripper(0.03)`
  How much space is there ultimately between the fingers?

5. Add a function call to move to a desired end-effector pose:
```
des_pose= RigidTransform(rotation=np.array([
        [1.0, 0.0, 0.0],
        [0.0, -1.0, 0.0],
        [0.0, 0.0, -1.0]]),
        translation=np.array([0.3, -0.2, 0.4]),
        from_frame='franka_tool', to_frame='world')
fa.goto_pose(des_pose, use_impedance=False)
```
   Note that the `use_impedance` is set to false, which means it uses the robot's internal Cartesian impedance controller. The rotation matrix and translation vector define the desired orientation and position of the end-effector frame relative to the robot's base frame. An overview of the robot arm's kinematics is given in Fig. 5. Which directions are x, y, and z for the robot? If you wanted to have the robot point its palm in the left direction, what would a suitable rotation matrix look like?

6. Adapt the function call to change the duration and adapt the Cartesian impedance gains (stiffness):
   `fa.goto_pose(des_pose,duration=10.0, use_impedance=False, cartesian_impedances=[3000, 3000, 100, 300, 300, 300])`
   You should now notice that the end effector is more compliant in the z-direction. Higher gains will result in better free-space tracking but can apply excessive forces in collisions. Very high gains can cause oscillations that are not good for the robot. Do not exceed 3000 and 300 for position and orientation gains, respectively.

Additional Exercise

Now that you are familiar with the basic operation of the robot, it is time for you to create your own script. Choose one of the following tasks (or multiple) to implement:

- Task 1: Pick up a block from a fixed location and stack it at another location. Can you stack multiple blocks?
- Task 2: Pick up a whiteboard eraser and slide it between two points on the table surface.

Remember to move slowly when placing objects to avoid high-speed collisions.

Further Reading

To learn more about frankapy, we recommend students look through the example scripts in frankapy/examples. Students should also explore the frankapy repository on github and its documentation at:
`https://iamlab-cmu.github.io/frankapy/index.html`

Turning off the Robot Arm

The robots are shared between students. As a courtesy to others, we ask that you always bring the robot back to its reset position and power it down properly after each session.

1. Reset the robot to its home position using the following commands:
   `bash terminal_docker.sh`
   `python3 reset_joints.py`
2. Close the terminals related to the Control PC to shut off the remote server.
3. Turn off the robot by returning to the google-chrome web interface on the Control PC and clicking shutdown under the menu tab, as shown in Fig. 4.
4. Wait! Wait for a minute for the robot to shutdown, then flip the physical switch on the FCI (do not flip the switch before the robot shutdown has finished).

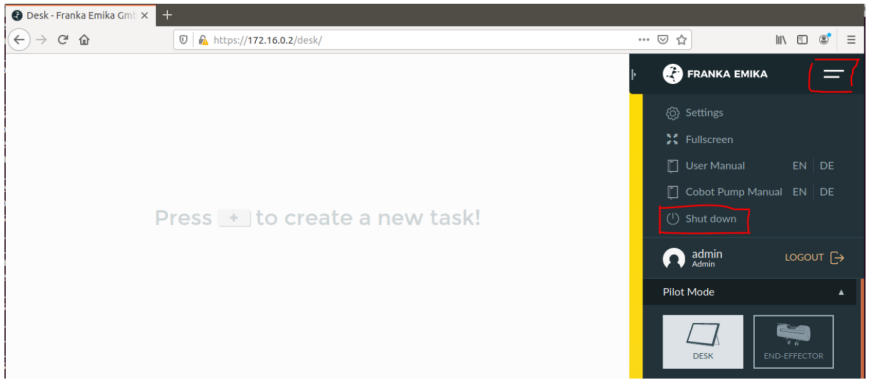

Figure 4: Location of the Shut Down button

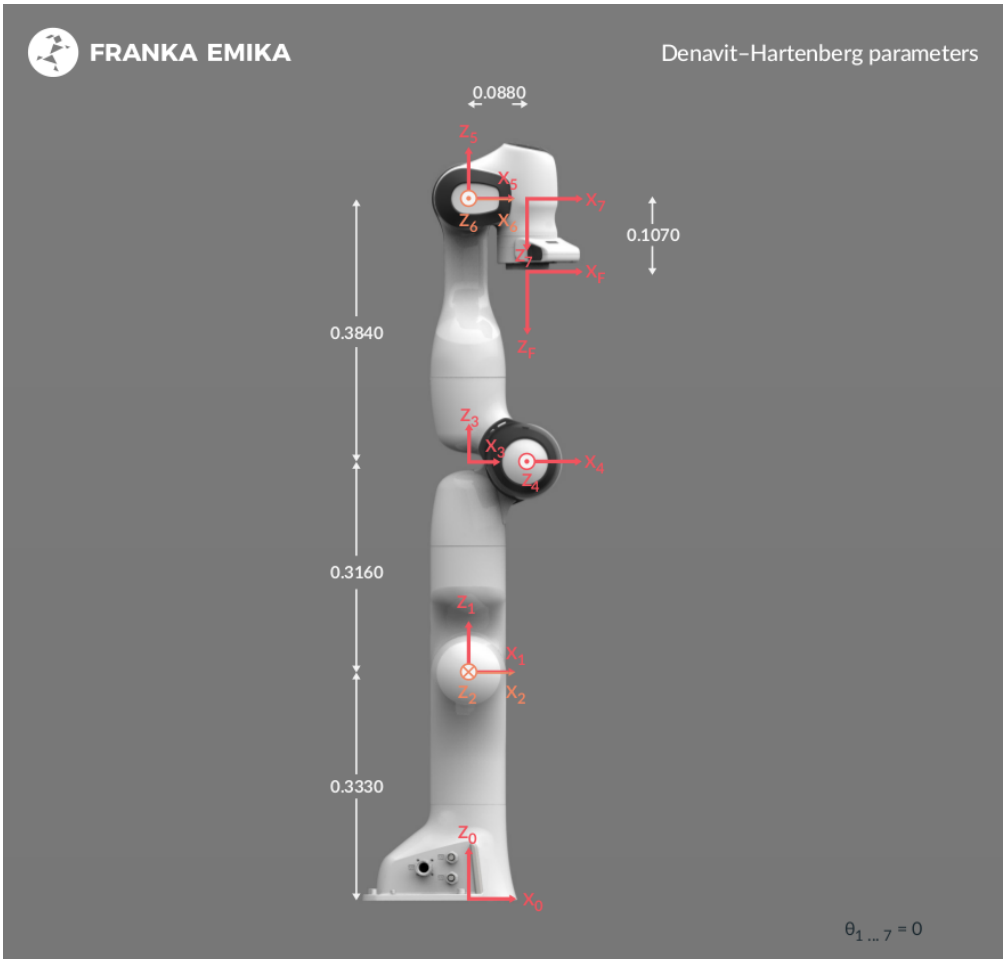

Figure 5: The coordinate frames along the robot's arm.